

Research

IoT and AI for smart agriculture in resource-constrained environments: challenges, opportunities and solutions

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Abstract

Climate change is an undeniable reality with far-reaching and profound implications for agriculture and, subsequently, global food security. The highly vulnerable farming communities in underdeveloped and developing countries can overcome resource and capacity constraints with the help of technology, particularly the Internet of Things (IoT) and Artificial Intelligence (AI), making agriculture smarter and more resilient to climate change. This work aims to enrich existing research on smart agriculture by focusing on its applicability in resource-constrained environments. In this context, it presents a detailed overview of the technologies enabling smart agriculture, highlights the challenges and opportunities for farming communities in developing countries, and proposes a framework for climate change-resilient smart agriculture. The framework is loosely based on McKinsey's 7S model for change management, consisting of hard and soft elements that are defined and adapted for the desired context. The hard elements include IoT sensors, network communications, and data management and analysis using AI, whereas the soft elements consist of policies and regulations, capacity building measures, and a supportive developmental ecosystem. This novel approach has not been employed before in this context. Furthermore, the framework's efficacy for environmental and crop growth monitoring is demonstrated through its implementation in a low-cost, open-source IoT system within a greenhouse using Edge-Cloud architecture. Here raw, extracted, and derived features are monitored to estimate irrigation requirements and crop maturity date. We conclude with an analysis of the results, recommendations for implementing climate change-resilient smart agriculture in resource-constrained environments, and the identification of areas for future research.

Article Highlights

1. Technical, financial, and social challenges in resource-constrained environments impede the adoption of smart agriculture practices.
2. A holistic framework, including hard and soft elements — as proposed in this work — enhances technology adoption and absorption rates.
3. Low-cost, open-source, IoT solutions are effective and affordable means for higher efficiency and productivity in agriculture.

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1 Introduction

1.1 Background

The global population is projected to reach nearly 10 billion by 2050, necessitating an almost doubling of agricultural production in developing countries [1]. This, combined with the challenges posed by climate change and limited capacity to modernize agriculture practices, presents a significant threat to meeting future food demands and achieving sustainability. Fortunately, the technological advancements of the twenty-first century offer promising solutions to help farming communities achieve sustainability by reducing input costs and increasing efficiency and yields [2–4].

1.2 Technology's role in smart agriculture

Smart agriculture refers to farming practices that incorporate state of the art sensing, computing and communication technologies to reduce input cost and wastage, increase efficiency and maximize yield. These are the three key tenants of sustainable farming practices as depicted in Fig. 1. These practices involve monitoring environmental, soil and crop health parameters, and making timely interventions through automated control of irrigation, fertilization and the application of pesticides and fungicides, in both open-field farming and controlled greenhouse environments.

Rapid advancements in sensing, computing, communications and intelligent systems—enabled by IoT, AI, big data analytics (BDA), blockchain, unmanned aerial vehicles (UAVs), and remote sensing (RS) technologies—have made remote monitoring, control, and data analysis capabilities readily accessible for smart agriculture [5–8] (Fig. 2). The large volume of multimodal data generated by agriculture IoT nodes enables forecasting and intelligent decision-making. This transforms traditional reactive farming into a highly efficient, proactive practice where inputs are applied only when needed and in the required quantities. This approach maximizes yield, reduces overall cost, increases productivity and mitigates adverse environmental impacts—all key traits of sustainable agriculture [9]. Real-time monitoring and control, data-driven intelligent and autonomous decision-making, and the ability to predict future events with greater accuracy are the key capabilities transforming agricultural landscape.

1.2.1 IoT in agriculture

IoT connects devices interfaced with physical systems to the cyber system—computational hardware and software located at a physical distance, accessed over communication network(s). Data from these devices is abstracted to knowledge and wisdom in the form of an actionable plan, through processing, storage and analysis. This capability can reside on the backend device, at network's Edge accessible over local area network (LAN), on remote servers in the Cloud accessible over the Internet or on any combination of these. IoT applications include monitoring and control, data visualization, and data analysis for advance planning and intelligent decision-making [10–14].

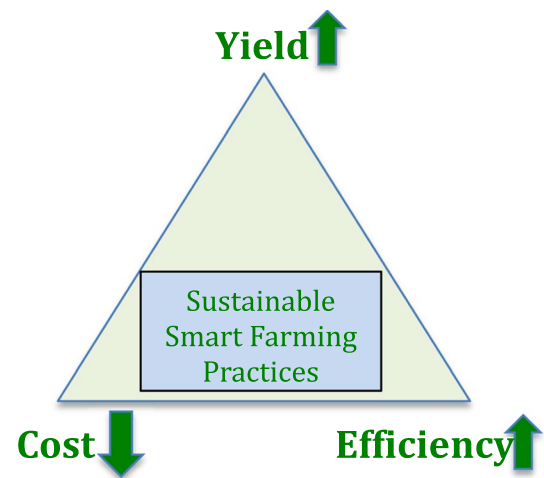
Given the emergence of food security as one of the major challenges of the twenty-first century, the world is increasingly turning to technologies like IoT and AI for higher operational efficiency and yields as 70% of the farming time is spent on monitoring and understanding field crops rather than performing the actual work [15, 16].

1.2.2 AI in agriculture

In 1956 John McCarthy coined the term Artificial Intelligence for the first time and defined it as a science and engineering of making intelligent computer programs [17]. Artificial Intelligence is an umbrella term used for a collection of technologies such as Machine Learning, Deep Learning, Big Data Analytics (BDA), Fuzzy Logic (FL), Computer Vision (CV), Expert Systems, Natural Language Processing (NLP) and Swarm Intelligence (SI), as shown in Fig. 3.

IoT systems generate large volumes of heterogeneous data at very high speeds, referred to as Big Data (BD), which is characterized by the 3Vs: volume, variety, and velocity [18]. The challenge then becomes how to efficiently and quickly extract meaningful information from this vast amount of data. The rapid proliferation of AI in recent years, enabled by advancements in Internet and Computing Technologies (ICT), offers a solution to this problem. AI models, learned from

Fig. 1 Key tenants of smart agriculture



data, not only reveal the relationships between inputs and outputs but also uncover hidden patterns and associations that may not be obvious or easily discoverable otherwise.

In arable agriculture, land suitability, sowing/planting time, irrigation and fertilization requirement forecasting, weed detection, disease detection or early warning, yield forecasting, weather forecasting are some of the areas where AI models have been effectively applied for intelligent decision-making [19].

1.3 Smart agriculture for resource-constrained environments

While much work has been done in the field of smart agriculture, its application in resource-constrained environments has received less attention. This work aims to provide a comprehensive survey of the current landscape of smart agriculture within the context of resource-constrained environments, identify challenges and opportunities, and propose a novel framework for its integration in farming communities of the developing countries. To the best of our knowledge, this is the first work of its kind in this specific context.

Fig. 2 Key enabling technologies for smart agriculture

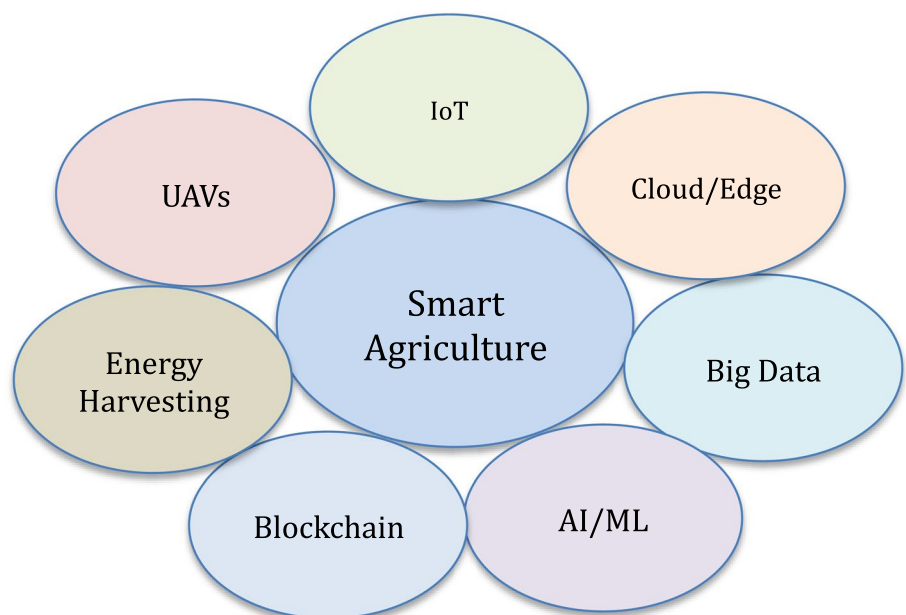
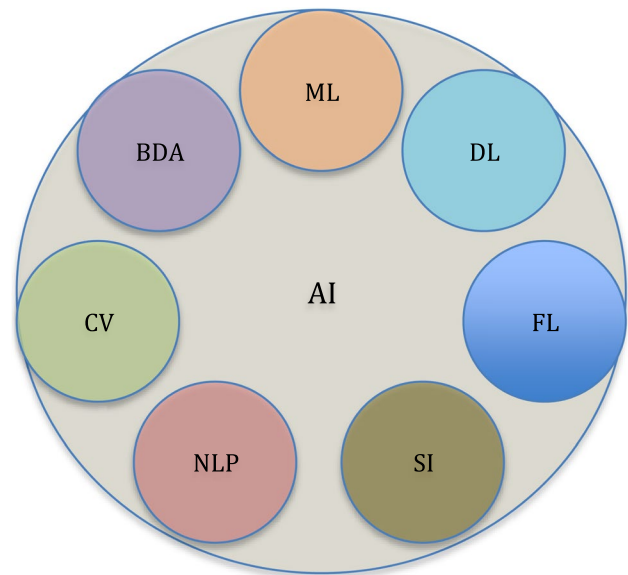


Fig. 3 AI encompassing technologies



2 Systematic literature review of IoT and AI for smart agriculture

2.1 IoT for smart agriculture

2.1.1 Sensing

In smart agriculture, environmental, soil and plant health parameters are monitored using various types of sensors. Sensors are transducers that convert sensed quantities into electrical signals. A sensor-actuator pair provides a closed-loop control system in which, a predetermined action can be taken based on certain threshold of sensed quantity. Sensors can be analog or digital depending on the type of output [19].

Recent advances in technology have enabled integration of processing and communication capabilities in sensors providing an evolutionary path from plain old M2M to IoT and beyond. This has opened up a whole new set of possibilities for networking and computing. Ermanno Cordelli, et al. proposed an open-source flexible software-based architecture for smart sensor that can be used in IoT applications [20]. This proposed reference architecture, which consists of a sensing interface, measurement engine and communication modules is configurable as against fixed COTS system, which is generally static and not upgradable.

When it comes to resource-constrained agricultural environments, cost, reliability, and energy consumption are the key constraints. Robles et al. proposed a low-cost and energy-efficient self-charging cloud-based IoT system for small and medium rural agriculture farms [21]. They created several prototypes and demonstrated their efficacy for the purpose of monitoring and control of irrigation by installing them in a sensor network connected to the cloud with self-charging capability. Rosa et al. proposed a battery-free energy harvesting wireless smart sensor platform consisting of low-cost off the shelf components capable of Bluetooth Low Energy (BLE) connectivity for smart agriculture and demonstrated that it can effectively communicate within a radius of 160 m even in low light-intensity environments such as vineyards [22]. Ferrer-Cid et al. collected raw data for air pollution with low-cost electrochemical sensors. Their sensor nodes consisted of NO₂, O₃ and NO sensors and sampled data at a high frequency of 0.5 Hz, which is important for analysis of time-series data for signal processing [23].

Monitoring of crops and fields through imaging is an area of great interest in research and much work has been done in this regard. Application of Field Programmable Gate Arrays (FPGAs) for real time imagery has been studied extensively and Antora et al. demonstrated its efficacy with an FPGA-based Image Processing (FIP) system for agricultural field monitoring application using a digital single-lens reflex (DSLR) and webcam both in the laboratory and field environment. Table 1 summarizes different types of sensors used in agriculture with their working principles and applications [24].

Table 1 Agriculture sensors types, working principles and applications

Categories	Sensing parameters	Mechanism	Types	Applications
Environment	Temperature	Electrical	Resistive, Capacitive, Inductive, Eddy Current	Irrigation Management, Growth conditions, Disease Susceptibility
	Humidity	Electrical, Optical	Resistive, Capacitive	
	Pressure	Electrical, Electromechanical	Capacitive, Piezoelectric	
	CO2, vapor, gases	Electrochemical, Optical	Resistive, infrared (NDIR)	
	Light Intensity	Photoelectric	Resistive	
	Wind Velocity	Electromechanical	Vane, thermal and cup anemometers	
	Moisture	Electrical	Tensiometers, Capacitive, Resistive, Hygrometer	
Soil	Temperature	Electrical	Resistive, Capacitive, Inductive, Eddy Current	Irrigation Management
	pH, NPK, salinity	Electrochemical	Electrical Conductivity	
	Soil Structure	Electromagnetic, Optical, Acoustic, Mechanical	Capacitive, Reflectance, Ultrasonic, Mechanical Resistance	
Crop/Plant	Weed	Acoustic/Optoelectrical	Ultrasonic, optoelectric	Weed Detection
	Yield	Electromechanical	Mass-Flow	Yield prediction
	Leaf Moisture	Optoelectrical	RGB and infrared imaging	Irrigation Management
	Health	Optoelectrical	Imaging sensors/cameras	Growth/Disease Detection

2.1.2 Communication technologies & protocols

The IoT Network Layer is responsible for the communication of sensed data to cloud and intermediary platforms over the Internet or Local Area Networks through wired or wireless communication [25]. At the sensor level, Wireless Sensor Networks (WSNs) are employed for communication among sensors and devices and between sensors and gateways or proxy servers. IoT nodes deployed within close proximity of each other i.e. a few meters, use low-power Personal Area Networks communication technologies such as Bluetooth, RFIC, NFC, and LowPAN, to transfer data or communicate within the network [12].

For IoT communication networks spanning over area up to 100 m, Local Area Network communication technologies such as WiFi, Bluetooth Low-Energy (BLE), and Zigbee are employed for wireless communications. For larger geographical areas spanning over several kilometers, cellular data networks such as GPRS/LTE/4G/5G, Narrow Band IoT (NB-IoT) or specifically designed narrow-band, long-range and low-power communication technologies such as Long Range Wide Area Network (LoRaWAN) and SigFox are employed [15]. The selection of a particular technology for networking depends not only on its range but also energy consumption, bandwidth, latency and robustness to errors and noise. A comparison of various low (< 10 m), medium (≤ 100) and high (> 100 m) range wireless technologies is provided in Table 2.

The IoT middleware layer ensures interoperability of heterogeneous devices by managing all IoT sensor nodes and their associations. It provides protocols to forward data from the sensor nodes or Edge devices to the cloud servers. A variety of protocols are offered by this layer to efficiently manage the data transfer process for a variety of scenarios based on software and hardware used, including MQTT, CoAP, XMPP, AMQP and DDS [13, 26].

2.1.3 Computing

Cloud Computing, Fog Computing, Edge Computing and Mist Computing are computing paradigms for IoT that define where the computation, storage and analysis capabilities reside in the network from sensor nodes to the applications and services hosting infrastructure. In Cloud Computing, software, platform and infrastructure are offered as services through private Cloud, Community Cloud, Public Cloud and hybrid Cloud. Software-as-a-Service (SAAS), Platform-as-a-Service (PAAS) and Infrastructure-as-a-Service are service models used under Cloud Computing for offering virtualized, pooled and shared resources for data storage, analysis, visualization, forecasting and intelligent decision-making [27–29].

Cloud-based data collection, analysis and prediction of farm activities through virtualization of farm resources utilizing Farm-as-a-Service (FaaS) model, enables crop growth monitoring and disease detection. Utilizing this concept Kim, et al. proposed FaaS integrated system, using an IoT-Hub model consisting of three layers: the network layer, FaaS layer and Application layer for strawberry disease prediction [30]. Using General Infection Model for plant disease occurrence probability, their results suggest that the disease onset temperature for the specific plant under consideration is around 20 °C. The disease susceptibility decreases significantly in temperature range of 15–20 °C and increases moderately between 20 and 30 °C. It is also observed that higher humidity increases disease spread but if wetness of leaves is minimized through continuous drying, the chances of disease spread are minimized. Sudden drop in nighttime temperatures also increases disease occurrence.

The Fog Computing paradigm brings the storage, analysis and networking capabilities of the Cloud a step closer to the IoT end of the network, typically located between the Edge and the Cloud [31, 32]. The difference between Edge and

Table 2 Comparison of low, medium, and high range communication technologies

	Frequency band	Data rate	Range	Energy consumption	Cost
WiFi	5–60 GHz	High	Medium	High	High
WiMAX	2–66 GHz	High	High	Medium	High
LR-WPAN	868/915 MHz, 2.4 GHz	Low	Low	Low	Low
Cellular	865 MHz, 2.4 GHz	Flexible	High	Medium	Medium
Bluetooth Low Energy	2.4 GHz	Medium	Medium	Very Low	Low
LoRa	868/900 MHz	Low	High	Very Low	High
SigFox	868/902/920 MHz	Low	High	Very Low	Medium
NB-IoT	800/900/1800 MHz (EU)	Low	High	Low	High
Zigbee	2400/915/868 MHz	Medium	Low	Low	Low

Fog Computing is the limited capabilities of Edge Computing due to higher comparative resource constraints of edge devices. Iván Froiz-Míguez, et al. designed and implemented an IoT smart irrigation system based on LoRaWAN to cover large urban areas for the collection of soil and air temperature/humidity data for either autonomous activation of water supply through Fog Computing gateways or through remote commands from cloud [33]. In their approach, a three-layer architecture is proposed consisting of IoT Sensing layer (LoRaWAN transceivers, sensors and activators), Fog Computing layer (edge devices and local servers), and Remote Service layer (Cloud). The 433 MHz LoRaWAN frequency is observed to provide better performance due to higher transmit power and lower radio propagation losses under various network configurations for gateway placement. A 23% reduction in water use is estimated just from considering weather forecast in irrigation decision-making.

Edge Computing brings the computation, storage and analysis capabilities of the Cloud to the network edge within close proximity of the IoT devices [34, 35]. Heterogeneous data collection in agricultural WSNs suffers from redundancy and latency, and the high volume and velocity of data results in poor response time for critical events. Edge-assisted data collection can overcome these problems by optimizing number of sensors required, reducing the volume of data, reducing latency and removing redundancy. Xiaomin Li et al. demonstrated this in [36] where they proposed an edge computing (EC) enabled data collection approach for critical events (CE) in smart agriculture for a software-defined WSN (SDWSN). In this scheme, Key Features Data Types (KFDTs) were extracted from historical data and used to define the collection data types in SDWSN. The proposed flexible framework consisting of WSN, SDWSN, Edge and Application layers dynamically adjusts the network or sensing parameters using three algorithms based on Mutual Information (MI), event identification based on minimum variance and sensing methods with time constraints on SDN.

Mist Computing is the other extreme end of the computing that resides on the IoT sensor nodes. The IoT sensor nodes are extremely resource-constrained devices with limited storage and processing capabilities mostly consisting of micro-controllers. These Mist nodes provide basic preprocessing capabilities before passing the data on to Edge or Fog nodes [37]. Comparison of IoT computing methods is given in Table 3.

2.2 AI applications in agriculture

2.2.1 Soil nutrition

Soil characteristics play an important role in determining the suitability of land for particular crop types. Soil type, texture, composition, and macro and micro nutritional content are some of the key parameters of interest in this context. Duraj et al. proposed an expert system consisting of IoT sensors and a Multi-Layer Perceptron (MLP) neural network to recommend suitability of land for cultivation [38]. Data collected through various sensors including soil moisture, salinity, pH and electromagnetic sensors is used to train MLP and it is determined that MLP with four hidden layers provides effective results for multiclass classification results compared to other models. Beside sensor data, some data related to soil structure and compactness etc. is obtained through manual surveys.

2.2.2 Irrigation management

Irrigation management is one of the most widely studied and demonstrated applications of IoT and AI in agriculture due to its significant practical implications. Efficient irrigation systems, such as drip irrigation—a low-pressure system, coupled with IoT and Variable Rate Technology (VRT)—provide significant savings in water usage. Abioye et al. has proposed and implemented an IoT monitoring framework to collect soil temp/humidity, air temp/humidity, and weather and irrigation data. Reference Evapotranspiration (ET_0) computed along with measured soil moisture data are used to schedule drip irrigation for growing mustard leaf plant in a green house [39]. Collected data and images are forwarded to the cloud through gateway, where data is available for storage and display. Comparison of different models for soil moisture content reveals that ARX model is more accurate (91.30%).

Torres-Sanchez et al. proposed an Irrigation Decision Support System (IDSS) taking into account the climate and soil data for citrus orchards in Spain [40]. Uses three regression models including Linear Regression (LR), Random Forest Regressor (RFR) and Support Vector Regressor (SVR). In case of SVR both linear and non-linear functions were considered and a Radial Basis Function (RBF) kernel was used for its better performance in non-linear relations. RFR outperformed other models in predicting the irrigation requirements through various stages of crop development.

González-Teruel et al. used soil and weather time-series data applying supervised binary classification and regression approaches with Random Forest and Support Vector Machine algorithms to predict water stress in mature cherry trees

Table 3 Cloud, Fog, Edge, and Mist computing comparison

Features	Computing Paradigm			
	Cloud	Fog	Edge	Mist
Location	Remote Data Center accessible over Internet	Near network edge between IoT node and Cloud	Network edge	IoT Sensor Node
Capabilities	Store, process, analysis	Limited store, process, analysis	More limited store, process, analysis	Pre-processing
Resources	Unlimited	Limited	More limited	Extremely limited
Strengths	Scalable, extendible, flexible	Real-time, low latency, limited offline	Low latency, energy & bandwidth, security, offline	Real-time, very low latency, offline
Weaknesses	High bandwidth & latency, security	Limited storage & computation	Limited storage & computation	Extremely limited

[41]. Majority Weighted Minority Oversampling Technique (MWMOTE) was applied to balance the dataset, which contained fewer samples of minority class. Their results suggest limited contribution of information from soil moisture sensors in regression algorithms. Similarly, Dubois et al. applied data driven Machine Learning approach for weekly prediction of Soil Water Potential (SWP) at three different depths in potato crop using supervised machine learning algorithms (NN, RF, and SVM) [42]. They conducted experiments with data acquired over 3-year period employing feature selection and using MAE, RMSE and R squared for performance evaluation. ETo is a crucial factor for determining irrigation requirements and its accurate measurement results in highly efficient water usage. Nawaz et al. proposed a hybrid ensemble algorithm for estimation of ETo using meteorological time-series data spanning over 5 years and showed that it outperformed six other regression ML algorithms with varying number of attributes in datasets [43].

Using open-source hardware and software, Goap et al. proposed an irrigation requirement prediction system [44]. Sensed parameters included soil moisture and temperature, air temperature and humidity, and UV radiation. Weather data was also incorporated in decision-making process. They used k-means clustering algorithm with a Support Vector Regressor model to predict Soil Moisture Differences (SMD). Similarly, Khriji et al. also implemented a Precision Agriculture system for temperature and soil moisture monitoring using open-source hardware and software [45]. Zia et al. demonstrated the efficacy of an IoT-based irrigation system by providing an experimental comparison of IoT-based and traditional irrigation on a flood-irrigated subtropical lemon farm [46]. Similarly, Mohammed et al. proposed an efficient IoT-based control system for a smart subsurface irrigation system to enhance irrigation management of a date palm orchid [47].

2.2.3 Crop health & disease detection

Monitoring of crop growth and early detection of disease is crucial for enhanced profitability. In this regard, embedded AI, an emerging field in the domain of AI enabled by rapid advancements in the processing capabilities of Microcontroller Units (MCUs) and external Graphics Processing Units (GPUs), has an important role [48]. Dimitrii Shadrin, et al. demonstrated the robustness of LSTM compared with other simpler approaches for plant growth modeling using AI-based embedded system model [49].

Multimodal data fusion is an approach that has been successfully used by many researchers to determine optimal growing conditions. Rezvani, et al. employed IoT sensors, LoRaWAN communications and Cloud computing technologies to create an experimental setup that recorded variations in temporal and spatial distribution of microclimate parameters (Temperature, Humidity and Vapor Pressure Deficit), from required optimum values, in a greenhouse used for commercial production of Tomatoes [50]. Employing sensor fusion with Optimality Degree (OptDeg) membership function, microclimate parameters are assessed at five growth stages of tomato crop. The membership function is mapped to a value between 0 and 1 that quantifies its optimality for tomato growth at a particular stage with a value closer to 1 indicating optimum conditions. Munnaf et al. proposed a site-specific seeding (SSS) approach for potato crop based on multi-sensor data fusion with k-means clustering that included sensors for various macro and micro soil nutrients along with soil pH, moisture content and Normalized Difference Vegetation Index (NDVI), which was measured with an NIR reflectance spectroscopy sensor [51]. When compared with Uniform Rate Seeding (URS) their approach showed significant improvement for tuber yields.

The disease susceptibility of a plant is quantified by considering pathogens (virulence and density), environmental conditions (temperature, humidity, etc.) and the host plants and their interactions with pathogens (contact, disease onset conditions) [30]. Paymode et al. proposed improvement in Multi-Crop Leaf Disease (MCLD) detection through image classification by pre-trained CNN-based Visual Geometry Group (VGG16) model using PlantVillage dataset from Penn State University [52]. With a performance achievement of 98.40% accuracy in grapes and 95.71% in tomatoes, the proposed model claims to have achieved greater accuracy for the classification of diseases. Similarly, Ahmed et al. proposed an ML-powered mobile-based automated plant leaf disease detection and classification model using Convolutional Neural Network (CNN) for resource-constrained farmers [53]. They achieved an accuracy of 94% for 38 categories of diseases from 15 crop species. Kamal et al. demonstrated the efficacy of depthwise separable Convolution Networks for diseases detection in comparison with conventional CNNs [54]. They trained and tested several models with PlantVillage dataset and discovered that Reduced MobileNet achieved an accuracy of 98.34% with 29 times fewer parameters than VGG and 6 times fewer than MobileNet on training data. The comparable accuracy of separable CNN models with conventional CNN models with significantly lesser parameters makes them suitable for resource-constrained IoT devices. Garcia et al. employed Near-Infrared (NIR) imaging cameras installed on an Unmanned Aerial Vehicle (UAV) to detect fungal disease

in cornfield [55]. They used Transfer Learning (TL)-based CNN and achieved an accuracy and precision of 86.7 and 98% respectively, demonstrating the efficacy of NIR for disease detection in corn.

2.2.4 Yield estimation

Another useful application of IoT and AI in agriculture is yield estimation. Peng et al. used satellite-based Solar-Induced Chlorophyll Fluorescence (SIF) dataset for predicting maize and soybean yield with SVM, ANN and RF algorithms, which outperformed LASSO and RIDGE regression algorithms [56]. E. Kamir et al. used a combination of satellite image, climate records and Machine Learning models to predict the estimated wheat yields in Australia [3]. K. Kuwata et al. employed deep learning models with remotely sensed data to estimate crop yields [57]. Ikram et al. proposed an ensemble algorithm consisting of Decision Tree, KNN, SVM, Random Forest and Gaussian Naïve Bayes ML algorithms and trained it on soil and meteorological data to predict 11 different crops with an accuracy of 97–98% [58].

2.2.5 Greenhouse management

Ge et al. studied the impact of eight environmental factors in a greenhouse on the estimation of Evapotranspiration [59]. They used an XGBR-based ET prediction model for drip-irrigated greenhouse and compared it with seven other regression models. The XGBR-ET performed well in modeling the daily ET in a greenhouse. They argued that the Penman–Monteith model for ET does not accurately model in a greenhouse environment and hence the need for hybrid ET estimation models for greenhouse crops. Sagheer et al. implemented a multi-tier cloud-based IoT system for monitoring and control of cucumber crop in soilless medium in a greenhouse and demonstrated increase in crop yield and savings in electricity and water consumption proving the efficacy of the system [60].

Silke Hemming et al. carried a benchmarking experiment in greenhouse environment with competing control strategies based on AI, assessing the impact of several environmental and crop parameters on crop yield [61]. Environmental and crop parameters were monitored and controlled through onsite sensors and actuators and decisions for lighting, ventilation, irrigation and fertigation were made based on AI algorithms. It was determined that three most important factors in crop yield are the cumulative duration of light (natural and artificial), temperature and the concentration of CO₂ through various stages of crop growth. This experiment, however, utilized synthetic datasets for the training of competing AI algorithms based on KASPRO greenhouse climate model since real datasets were not available.

System Identification is a modeling technique used to create representation of complex dynamic systems using different model structures. By linearizing and reducing model complexity it brings to surface the dominant dynamic modes of the system. Abioye et al. implemented an IoT-monitoring system in a greenhouse, measuring environmental, soil moisture content, irrigation volume and reference Evapotranspiration (ET_o) and compared ML algorithms (Auto Regressive with External Input—ARX, Auto Regressive Moving Average with External Input—ARMX, Box Jenkins—BJ) in MATLAB for prediction of irrigation schedule [39]. They employed System Identification method of modeling for the variables of drip irrigation system.

2.2.6 Machine learning algorithms and their applications in smart agriculture

Supervised, unsupervised and reinforcement machine learning algorithms are used in agriculture with applications ranging from crop selection based on soil analysis and climate to crop harvesting and yield estimation. Environmental and soil parameters are predicted to facilitate irrigation management that involves forecasting temperature, humidity and soil parameters such as moisture and temperature. Crop health management is carried out through weed and disease detection and proactive assessment of fertilization requirements. A list of most commonly used ML algorithms in agricultural applications are provided in Table 4.

2.3 UAV's in agriculture

UAV's are finding an increasing role in multiple segments of society. Their role for data collection and performing certain tasks in agriculture has been discussed in the literature in detail. In order to achieve fast, energy-efficient and robust data collection in a Wireless Sensor Network (WSN) deployed for smart agriculture, Dimitrios, et al. has proposed a network

Table 4 ML algorithms and their applications in smart agriculture

Algorithm	Soil suitability	Irrigation control	Health Mgmt	Yield estimation
ELM	✓			
ELM-R	✓			
RELM	✓			
Random Forest	✓			✓
Cubist	✓			
XGBoost	✓			
SVM	✓			✓
LS-SVM	✓			
ANN		✓		✓
BPNN		✓		✓
MLR		✓		✓
GB Regression		✓		
RF Regression		✓		
PLSR		✓		
ANFIS		✓		
CNN			✓	
RNN				✓
kNN				✓
Decision Tree				✓
LASSO				✓
RIDGE				✓

architecture that uses drones as mobile gateways (sinks) that fly over the remote sensor nodes and collect sensed data using a combination of long range low bandwidth (LoRa) and high bandwidth short range (WiFi) protocols [62]. The field nodes communicate with the Command Center (CC) using LoRaWAN protocol. Limited scale experimental results in lab demonstrated promising results in terms of reliability and energy efficiency.

Zhang et al. proposed use of UAVs (drones) for data collection as an energy-efficient technique to prolong the network lifetime. Joint optimization of nodes' wake-up schedule and drone's trajectory is undertaken to minimize the maximum energy consumption of all SNs while ensuring that the required amount of data is collected reliably from each node [63]. Similarly, Mahdi et al. investigated usage of drones for energy efficient data collection with specific emphasis on minimizing energy consumption of drone while collecting data in addition to the energy consumption from data transmission [64]. They employ clustering technique and travel salesman solution to optimize the drone trajectory for efficient data collection.

3 Smart agriculture for resource constrained environments—challenges & opportunities

In any well-established system, introducing changes to traditional practices and conventional methods through adoption of new technologies faces several challenges. Agriculture is no exception [65]. The inertia and inherent inefficiencies of the system, along with the lack of capacity of stakeholders, are key impediments to change. This is even more pronounced in the developing countries, where technological, economical and social challenges for adoption of smart technologies are enormous. Farming communities are often ill-equipped, both technically and financially, and have little incentive and support to experiment or innovate. The challenges for IoT and AI application in smart agriculture within resource-constrained environments can be categorized as financial, technical, and social, as summarized below.

3.1 Financial

1. **Affordability**—in developing countries, majority of the farmers are small-farm holders with very low incomes and most struggling to stay afloat. Technology interventions or any modernization efforts such as IoT and AI are cost-prohibitive.
2. **Market risk**—volatility in the prices of inputs and outputs due to various reasons including extreme weather swings caused by climate change make the farmers apprehensive about any new expenditure. The payoffs in the presence of volatile market conditions become uncertain thereby increasing resistance to invest in modernization or upgradation.
3. **Financial Inclusion**—there is a lack of concerted effort at the government level to introduce financial mechanisms and instruments for small-farm holders that encourage technology adoption.
4. **Financial Incentives**—without clear understanding and realization of the potential benefits that the technology stands to offer coupled with lack affordability and financial support, there is little incentive for struggling farmers to risk venturing into uncharted territories.

3.2 Technological

1. **Coverage**—agriculture happens mostly in remote rural areas where Internet coverage can be missing or spotty at best, making it difficult to transmit IoT sensor node data to the cloud servers.
2. **Bandwidth and latency**—deploying a large number of IoT nodes for farm monitoring across a region, with frequent data exchanges, will affect the network bandwidth and latency.
3. **Interoperability**—in the absence of universal standards, heterogeneous IoT nodes supporting various communication technologies and protocols when introduced in the network, cause interoperability issues.
4. **Power consumption**—IoT nodes deployed in the field are mostly battery operated and have limited lifetime, making power consumed by IoT sensor nodes a limiting factor in network lifetime.
5. **Heterogeneity**—Data generated by IoT nodes in an agricultural setup is highly heterogeneous, large in volume and generated at a very high speed (VVs).
6. **Noisy data**—the harsh environment in agriculture fields may cause the sensor nodes to be intermittent under adverse conditions i.e. these nodes may not be able to send data due to temporary power failures, communication failures or permanent equipment failures. Therefore, the data may contain missing and redundant values and outliers, resulting in bias, skew, and imbalance in dataset.
7. **Data privacy and security**—transmission of data through network on local premises and over the Internet exposes it to privacy and security concerns.
8. **Reliability**—cloud-based services such as Farm-as-a-Service (FaaS) become challenging in the absence of reliable Internet connectivity.
9. **Universal algorithms**—varying geographical conditions, parameters, and complex datasets pose a challenge for universal prediction & classification ML algorithms.
10. **Datasets**—lack of good quality location-specific training datasets for a variety of specific crops.
11. **ML limitations**—poor generalization of ML algorithms due to poor quality of training data.
12. **Computational costs**—high computational costs/resource requirements for data storage and running ML algorithms.

3.3 Social

1. **Education & skills**—in developing countries, farming is an occupation that runs in the family for generations where young ones are involved at a very early age leaving them without formal education and technical skills.
2. **Aversion to technology**—despite its shortcomings in the modern day and age, conventional wisdom prevails when it comes to farming practices and most small-farm farmers are skeptical to change due to lack of trust in technology.
3. **Lack of data**—farming sector is highly fragmented and there is little data available at farm, community and sector level in terms of land holding, yields, crops and incomes. This makes it difficult to create one-size-fit-all solutions.

4. Social Net—there is little cooperation and collaboration among farmers for technology adoption and the opportunities to do so are non-existent.
5. Adaptation to local context—tempo-spatial variations in farming conditions make it imperative to adapt technology to local context, make it customizable to individual farmer's needs and make it relevant and authentic. For example, mobile applications rendering information in local regional languages and graphics that are clearly understood by farmers.

3.4 Opportunities

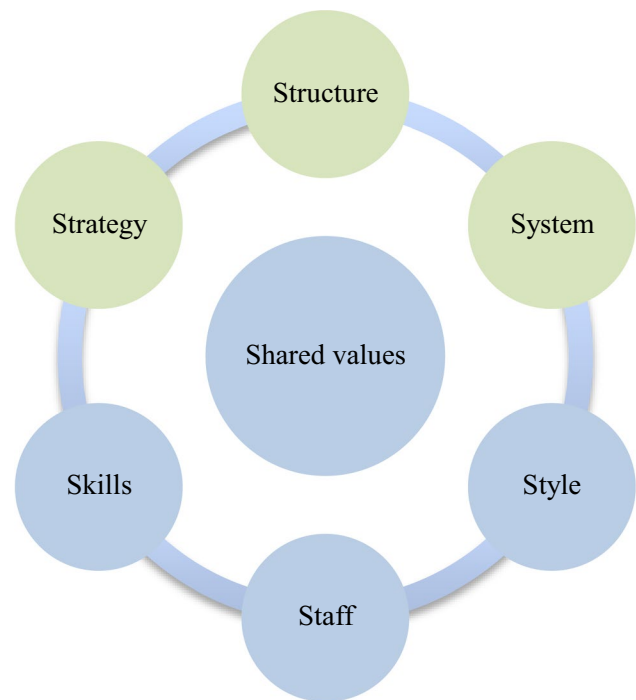
Despite the challenges stated above, the opportunities to increase income, productivity and efficiency through the application of IoT and AI technologies in smart agriculture are numerous. Adopting these technologies will not only prepare farmers for the uncertainty brought about by climate change but also protect their fields through sustainable practices. The opportunities offered by IoT and AI application in agriculture include:

1. Higher yields—continuous monitoring of fields and greenhouses ensures optimal growing conditions, leading to higher yields and, subsequently, higher incomes.
2. Lower costs—optimal use of inputs, enabled by continuous monitoring of environmental, soil and plant conditions through IoT nodes, results in greater efficiency, lower costs, and environmentally friendly, sustainable practices.
3. Productivity—autonomous, real-time monitoring and intelligent decision-making with IoT and AI free up farmers to focus on core business.
4. Virtualized services—virtualized farm services offered through Platform-as-a-Service and Farm-as-a-Service models saving farmers of high setup costs and steep learning curves.
5. Energy efficiency—energy harvesting, low-power consuming IoT sensor nodes improve network lifetime, ensuring smooth operations without the need of maintenance and manual interventions.
6. Localization—embedded AI enabled by smart IoT sensors and capable Edge devices allow for more powerful local data processing and analysis, without needing to send data to the cloud.
7. Prediction accuracy—machine learning models trained on location specific datasets improve relevance and prediction accuracy.
8. In-depth analysis and foresight—Big Data Analytics provides insights into phenomenon behind data, revealing hidden patterns and trends through visualizations that may not be observable otherwise.
9. Transfer learning—pre-trained models re-trained with minimal data are particularly useful for areas where little or no data is available for training ML models.
10. Federated Learning—allows collaborating client nodes to train a central model without having to share data, addressing privacy and security concerns while avoiding network congestion.
11. Continuous supply—IoT and AI controlled greenhouse farming ensures optimal growing conditions year-round, preventing shortages and overpricing.
12. UAVs—data collection and monitoring with UAV installed visual, IR/NIR, multispectral and hyperspectral cameras and sensors, swarm intelligence and autonomous spraying of pesticides and weedicides, ensuring healthy crops, higher yields, and low labor costs.

4 Framework for climate change-resilient smart agriculture in resource-constrained environments

The introduction of smart agriculture practices in resource-constrained environments requires a holistic approach. Given the complex nature of this task, it requires careful and thoughtful planning and execution, with clearly defined objectives, taking into account the constraints and limitations of the system and its stakeholders. We leverage the McKinsey's 7S framework design tool to come up with a conceptual framework that guides the introduction of smart agriculture in resource-constrained environments.

Fig. 4 McKinsey's 7S framework for change management



4.1 McKinsey's 7S framework for organizational change management

McKinsey's 7S framework is a multidimensional research tool originally designed for change management and assessment of organizational effectiveness [66]. It consists of seven elements: *strategy*, *system*, *structure*, *shared values*, *staff*, *style*, and *skills*, collectively referred to as 7S (Fig. 4). *Strategy* is a clear plan or course of action to achieve long-term goals, *Structure* is the organizational hierarchy and *System* is the processes and procedures that coordinate and govern the functioning of people and their daily activities. Together these three elements form the hard components of the framework. Whereas, the soft components of the framework include *Shared values*, which are the core values derived from vision and mission, beliefs, attitudes, and work ethics, *Style*, which is the management and leadership approach, communication style, and culture, *Staff*, which includes people with varying educational backgrounds, trainings and general capabilities, and *Skills*, which are the actual skills and competencies of people. The alignment and balance of these hard and soft components drive organizational performance towards achieving its objectives.

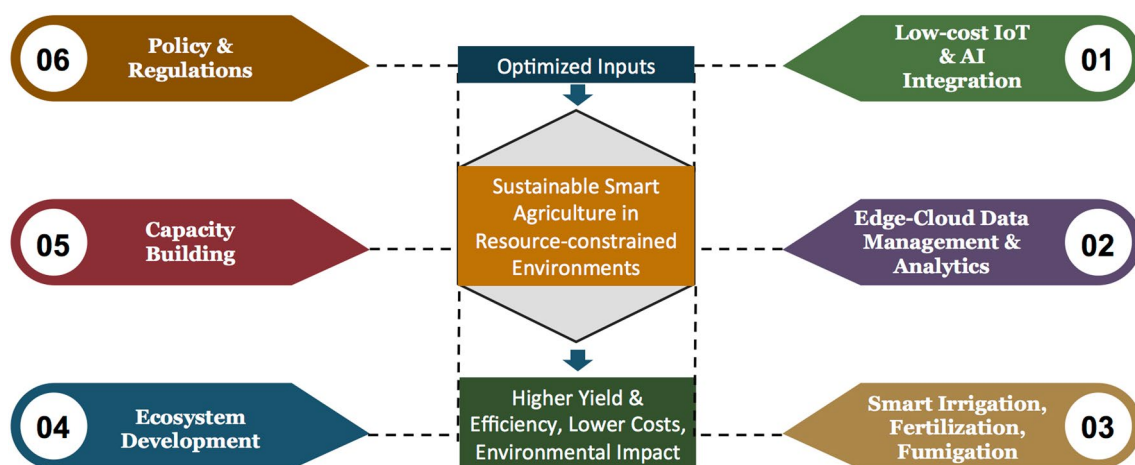


Fig. 5 Conceptual framework for integration of IoT and AI in agriculture in resource-constrained environments

4.2 Systemic change towards smart agriculture in resource-constrained environments—proposed framework

Our proposed conceptual framework, which is inspired by McKinsey's 7 s framework for change management is depicted in Fig. 5. This framework takes the central idea of McKinsey's 7 s framework, which is to identify hard and soft elements required for change, adapt these for agricultural context, and balance the two. This approach, grounded in balancing the hard and soft elements, aims to promote and support successful integration of sustainable smart agriculture in resource-constrained environments.

4.2.1 Hard elements of the proposed framework

The original framework is adapted by replacing the hard elements Strategy, Structure, and System with physical systems consisting of core smart agriculture technologies, including IoT sensors, network communications, and data management & analysis using AI. These core technology components form the basis of this framework. The hard components of this framework are further expanded in Fig. 6 to highlight requirements under resource-constrained environments. The areas focused are relevant data collection, communication, storage & analysis, applications, and services.

Low-cost IoT sensors put stringent requirements on data preprocessing, as these are prone to errors, missing values, outliers, and anomalies due to lower sensitivity and higher tolerances in signal measurements. Robust but lightweight filtering/smoothing, outlier/anomaly detection and subsequent imputation techniques are required to generate meaningful datasets for lightweight regression and classification ML algorithms for predictive analysis with low bias and variance. Data can also be obtained from other sources such as local weather stations and remote sensing to augment localized datasets. Low power and medium to long range communication technologies such as BLE for LAN and LoRa for WAN are suitable options. Edge computing supports preprocessing of data and lightweight ML algorithms for predictions and classifications. Irrigation, fertilization, and crop health management are the three primary applications, which can be offered as virtualized services through cloud platforms, such as FaaS model.

4.2.2 Soft elements of the proposed framework

In comparison with the soft elements of the original framework, we propose soft elements that support and enhance technology absorption through policy initiatives, capacity building, and development of affordable but effective

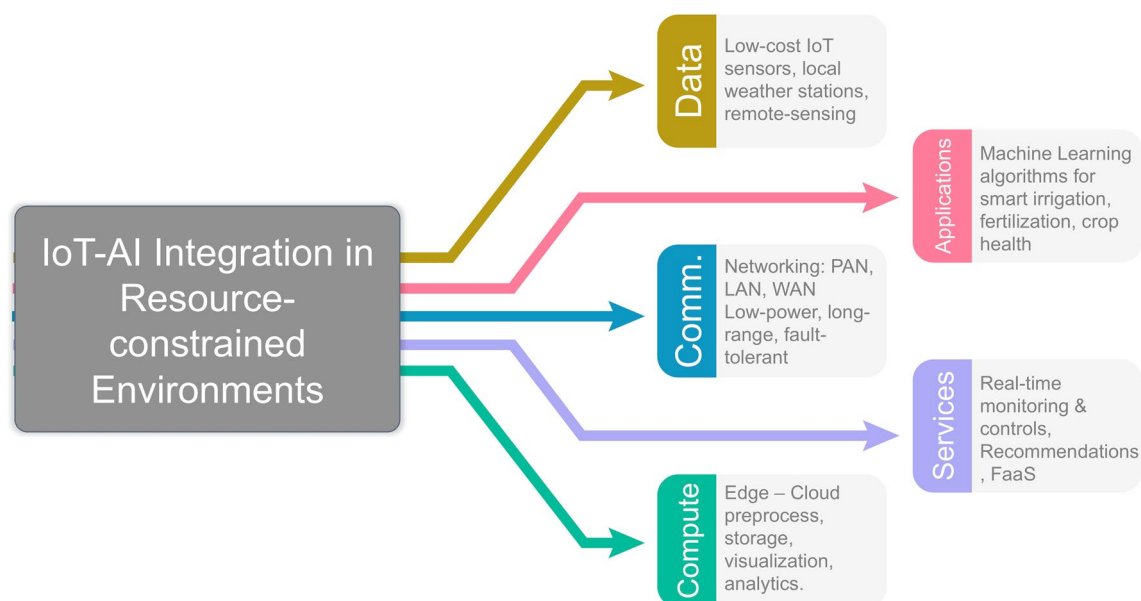


Fig. 6 Hard elements of the proposed smart agriculture framework for resource-constrained environments

Table 5 Soft elements of the proposed smart agriculture framework for resource-constrained environments

Soft Element	Requirements	Stakeholders	Outcomes
Policy	- Evidence-based - Research-informed - Progressive, Inclusive - Long-term - Farmer-centered	Government, Agriculture departments, Industry, Academia, Farming communities	Self-sustained policies enabling and promoting technology integration
Capacity	- Technical - Financial	Agriculture departments, Banks/Lenders, Industry, Academia, Farming communities	Skilled, tech savvy and financially secure farming communities
Ecosystem	- Public-private partners - Infrastructure - Solutions (products & services) - Supply chain - Tech support	Government, Telecom industry, Agriculture departments, Academia, Industry, Entrepreneurs, Incubators, Investors, Venture Capitalists, Farming communities	Modernization of agriculture practices through innovative approaches and solutions resulting in higher efficiency and productivity

ecosystem for climate change-resilient smart agriculture. The intended outcomes entail comprehensive strategies for smart agriculture integration in resource-constrained environments, based on well-informed and inclusive policies, promoting public–private partnerships to develop technical and financial capacity and a robust ecosystem to introduce, promote, and support upgradation of existing infrastructure and innovations in new products, services, and solutions. Table 5 summarizes these soft elements, their requirements, potential stakeholders, and intended outcomes.

5 Methods & materials

One of the goals of this research work is to propose a framework of affordable yet effective methods for IoT data collection, dataset creation, visualization and analysis for intelligent decision-making that are easily implementable in resource-constrained environments. A full-blown IoT setup at the farm level can be cost prohibitive and intimidating for new entrants in developing countries without having a realization and clear understanding of the benefits. Under these circumstances, creating and offering easily accessible services at very low or no cost can lower the barrier to entry.

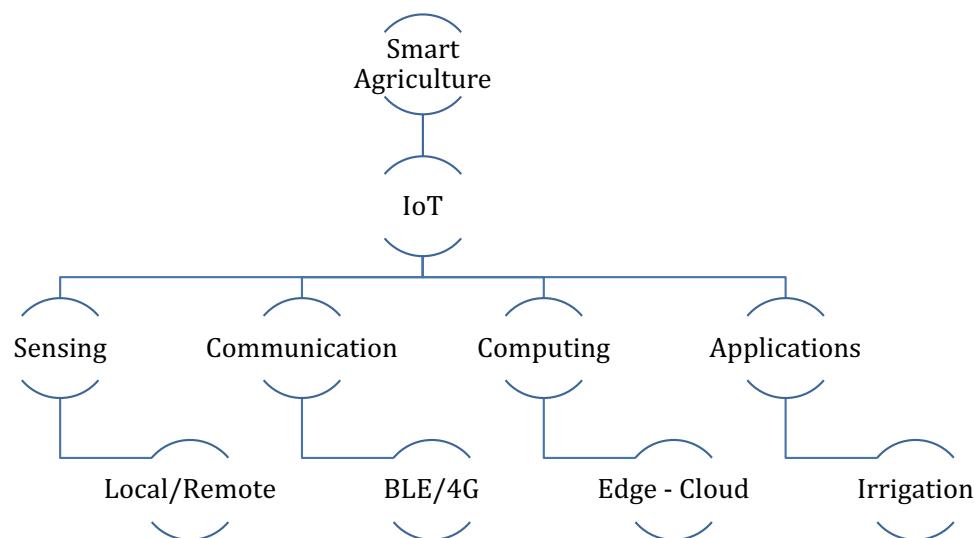
5.1 A low-cost open-source IoT enabled smart agriculture

A simplified proposed framework for implementation of smart agriculture for efficient and sustainable farming practices in small farms in rural areas consisting of sensing, communication, computing and applications is presented in Fig. 7. Data is generated by local in-field sensors, installed for continuous monitoring of environment and soil conditions including air temperature, relative humidity, atmospheric pressure, solar irradiance, soil temperature and soil moisture. This data is collected and preprocessed in the Edge before sending it to Cloud where it is available for visualization and further analysis using Machine Learning algorithms (Edge-Cloud hybrid approach). Since optimal use of freshwater is a key objective of smart agriculture, initially, we focus on irrigation management.

5.2 Proposed IoT architecture

A low-cost open-source end-to-end IoT architecture for this proposed framework consists of a sensing layer, network layer and application layer. In the sensing layer, low-cost temperature, humidity, soil moisture and irradiance sensors are connected to an affordable Arduino Nano Sense BLE 33 microcontroller unit (MCU). The collected sensor data is communicated to a Raspberry Pi 3B + Gateway/Bridge emulator over Bluetooth Low-Energy (BLE) protocol, for onward publication to the open-source ThingSpeak cloud server, using the IoT Message Queue Telemetry Transport (MQTT) protocol. This 3-layer architecture for greenhouse monitoring is depicted in Fig. 8.

Fig. 7 Proposed smart agriculture framework for implementation in small farms



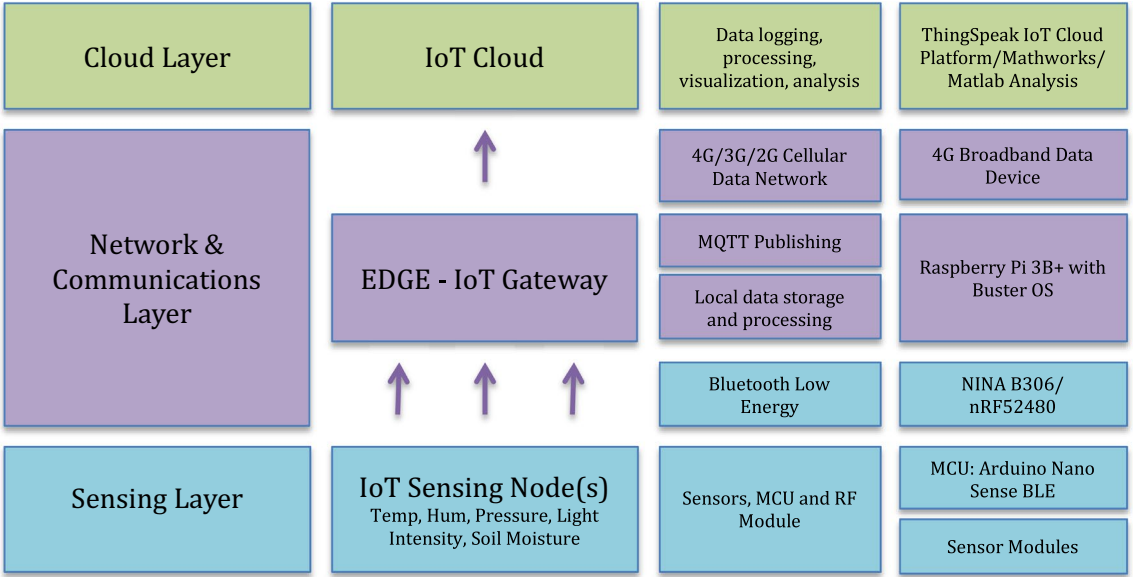


Fig. 8 Proposed 3-layer IoT architecture for greenhouse monitoring

Fig. 9 Information flow through IoT network

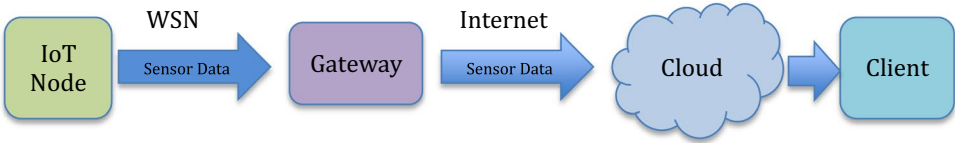


Fig. 10 Low-cost experimental greenhouse setup

The flow of information across the IoT setup, starting from the IoT sensor node and culminating at the cloud IoT server, for visualization and analysis accessible on client device, along with corresponding setup and processes at each step are presented in Fig. 9.



Fig. 11 Raspberry Pi 3B+ bridge, Arduino Nano 33 BLE MCU, and sensors setup

Table 6 Sensor specifications

Part	Sensor	Range	Accuracy	Interface	Supply volt
Inside temperature	HTS221	– 40 C to + 120 C	± 0.5%	SPI & I2C	1.7–3.6
Soil temperature	DS18B20	– 55 C to + 125 C	± 0.5%	1-Wire	3–5.5 V
Inside humidity	HTS221	0–100%	± 3.5%	SPI & I2C	1.7–3.6
Pressure	LPS22HB	260–1260 hPa		SPI & I2C	1.7–3.6
Soil moisture	CSOIL-Q205	0–100%	± 3.5%	AO/DO	3.3–12

5.3 Experimental setup

To demonstrate the validity and efficacy of this proposed framework, a low-cost setup is installed at a farm in Qadir Nagar, Buner, KP, Pakistan (Fig. 10). Two greenhouse structures made of locally available bamboo are erected and covered with polythene sheets. The length and width of each greenhouse is 10 feet and a height of 6 feet at the center as shown in Figs. 3, 4, 5, 6. Tomato seedlings are grown in mid-September and transplanted into the tunnel before mid-October. The tomatoes saplings are planted on raised beds 1.5 feet wide and approximately 12 inches high with one-foot wide furrows in between raised beds. The plant-to-plant distance is 1.5 feet and row-to-row distance is 2 feet. Each tunnel accommodates approximately 25 F1-Hybrid tomato plants. The irrigation is carried out using the conventional flooding method where the furrows are flooded with water.

5.3.1 Hardware, software and applications

We use a low cost Arduino Nano 33 BLE MCU with embedded sensors and Bluetooth Low-Energy RF communication module—a highly efficient protocol for LAN. This MCU is versatile and has very low power consumption (< 100 mA). Additionally, it runs Mbed OS that is capable of running ML models developed with TinyML framework—a framework for running ML models on resource-constrained devices. Low-cost external sensors for measuring air temperature and humidity, light intensity, soil temperature and soil moisture content are easily interfaced with this microcontroller. The sensed parameters are communicated over BLE to a Raspberry Pi acting as a local gateway/Edge device. The data can be stored onsite and preprocessed before publishing it through MQTT to an open-source IoT cloud services such as ThingSpeak for visualization and data analytics.

The BLE setup for communication between IoT Sensor Node and the Gateway/Bridge emulator requires creating Generic Attributes that include Profile, Service and Characteristics. Once created these generic attributes allow exchange of information between peripheral server and central client, which can be viewed on a mobile application for verification of the setup.

The IoT gateway functionality emulated on a Raspberry Pi 3B+ runs an RPi OS Buster (Linux-based). The Bluetooth Low-Energy (BLE) and The Message Queue Telemetry Transport (MQTT) protocols are implemented in gateway emulator using Python libraries. MQTT is a lightweight and efficient bi-directional (Client-Broker) message publishing protocol designed for resource constrained IoT devices. The connection between the client and the broker is supported over TCP

and Web Sockets with reliable message delivery and support for unreliable connectivity, which is a typical scenario for agriculture in remote and rural areas. The MQTT client side on Raspberry Pi 3B+ gateway emulator communicates with an online broker—ThingsView, an open-source cloud-based online service for data visualization and analysis. The gateway emulator/bridge, IoT node, and sensor setup is shown in Fig. 11. The specifications for sensors are provided in Table 6.

5.4 Data collection algorithm

Using this low-cost end-to-end open-source IoT setup, we collect environmental and soil data for real-time monitoring of the greenhouse, which includes inside air temperature and humidity, light intensity, soil temperature and soil moisture. The IoT node (Arduino Nano 33 BLE) from the field communicates sensor values to the gateway emulator (Raspberry Pi 3B+) over BLE. The data is logged in a comma-separated value (CSV) file on the gateway for preprocessing and subsequent transmission to MQTT broker over the Internet. We chose not to perform any preprocessing so that we can view raw data and qualitatively assess sensor performance. The data is visualized on the Thingspeak cloud platform. This platform also allows further data processing and analysis to derive useful insights for decision-making. Real-time data as well as the derived indicative crop growth parameters such as Growing Degree Days and Vapor Pressure Deficit are used for irrigation management, better understanding of crop developmental status, and optimal growth requirements. The algorithm used for data collection is as follows:

1. Deploy WSN in field.
 - i. Initialize Sensors.
 - ii. Initialize Gateway.
2. Repeat the following steps for the duration of crop cycle.
 - i. IoT Node reads sensor data at predefined intervals,
 - ii. IoT node defines BLE service & characteristics, package data in BLE packet.
 - iii. GW connects to IoT Node over BLE.
 - iv. GW reads data over BLE, stores locally in a CSV file.
 - v. GW connects to MQTT Broker, publishes to IoT ThingSpeak cloud service.
 - vi. ThingSpeak IoT service receives data, makes it available for visualization and analysis
 - vii. ThingSpeak IoT service generates alerts when user-defined limits are exceeded
 - viii. Exit when crop harvested.
3. End.

6 Results

6.1 Primary data collection

The data collected from installed sensors in the greenhouse included air temperature, relative humidity, soil temperature, soil moisture, irradiance and atmospheric pressure. The data was available for real-time monitoring on the ThingsView mobile application and ThingSpeak IoT cloud website as shown in Fig. 12.

6.2 Feature engineering

Besides raw sensor data providing instantaneous values, we created additional features by taking minimum, maximum and average values for each attribute. In addition to this we also derived two key indicative parameters for crop growth and irrigation management; Growing Degree Days (GDD) and Vapor Pressure Deficit (VPD) from primary sensed data. This way the dataset created by this setup includes primary sensed data, extracted features and derived features.

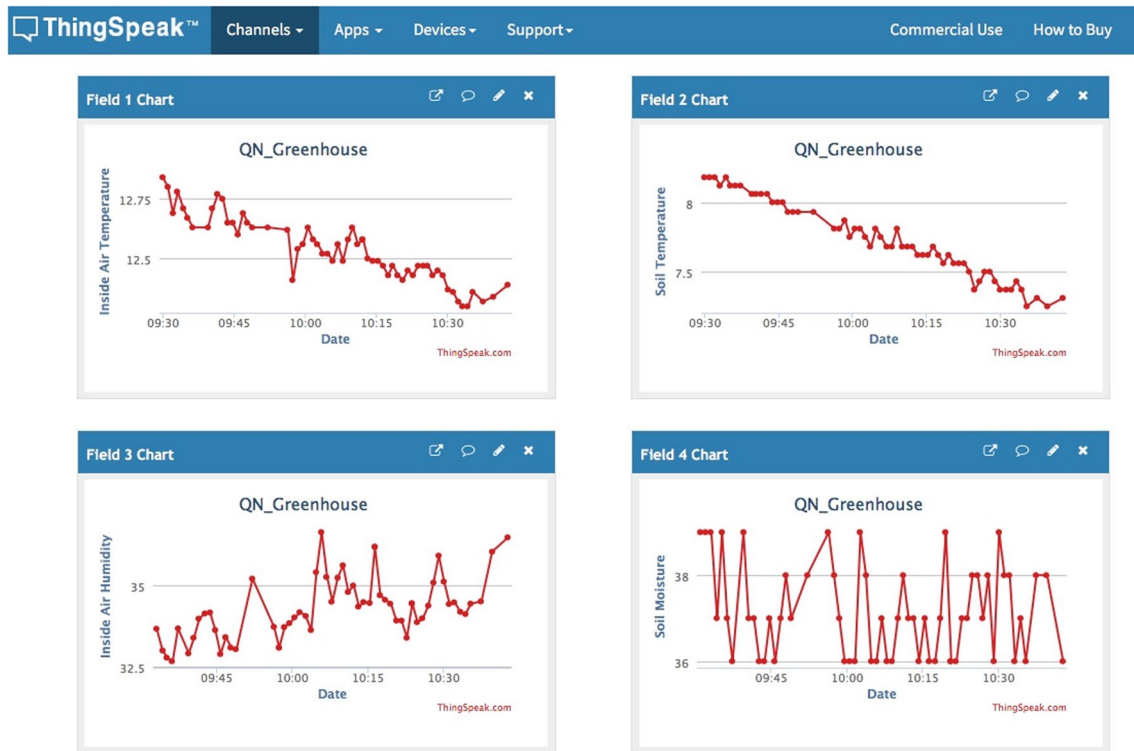


Fig. 12 Real-time visualization of IoT data from greenhouse on ThingSpeak website

6.2.1 Growing degree days—GDD

Plants need a certain amount of heat and light to grow. If this minimum threshold of light and heat is not exceeded, plants growth will be restricted and in some cases, stop completely. GDD, also referred to as Heat Units, is the accumulation of daily temperatures, taking in to account a minimum developmental threshold—a base temperature, which must be exceeded for the plant to grow [67]. The expression for GDD is given by:

$$GDD = T_{mean} - T_{base}, \text{ if } T_{mean} > T_{base} \quad (1)$$

$$GDD = 0, \text{ if } T_{mean} < T_{base}$$

where T_{base} is GDD base temperature.

and T_{mean} is mean temperature given by: $T_{mean} = (T_{min} + T_{max})/2$

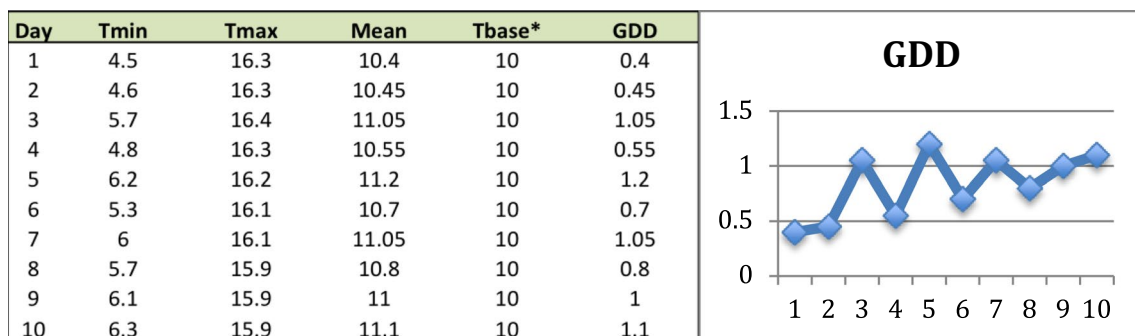


Fig. 13 GDD calculations for the greenhouse setup

GDD helps ascertain the plant growth stage proactively and retrospectively. Instead of relying on fixed calendar days, GDD helps farmers predict the likely crop maturity date. For tomato crop, typically this GDD value falls in the range of 1200–1800, depending on the variety. The base temperature for most crops including tomatoes is taken as 10 °C. Figure 13 shows calculation of GDD for the tomato crop in greenhouse over a period of 10 days. The cumulative GDD value for 10-day period is 8.3, with a mean daily GDD of 0.83, standard deviation 0.29, and variance 0.08. These values indicate a suboptimal growth period due to prevailing low temperatures and overcast conditions. The relatively low variability in GDD implies consistent environmental conditions affecting plant conditions during this period.

6.2.2 Vapor pressure deficit—VPD

VPD is a measure of the pressure that the water vapor exerts on the plant leaves. It is calculated as a difference between the amount of moisture in the air and the amount of moisture in the air when fully saturated. It helps determine optimal transpiration rate of plants and is an important factor for greenhouse growing [68]. VPD is inversely related to RH; higher VPD (lower RH) means less pressure exerted on leaves, which results in higher transpiration rate and loss of water. At the same time, lower VPD (high RH) means more pressure exerted on leaves resulting in lower transpiration rate, which may slow down plant growth and create conducive environment for fungal disease attack. VPD is expressed as:

$$VPD = e_s - e_a = e_s - (Rh * e_s / 100) \quad (2)$$

where e_s is the saturated vapor pressure.

e_a is the actual vapor pressure

and Rh is relative humidity

$$e_s(\text{millibars}) = 6.1078 \exp\left(\frac{(17.269 * T)}{(237.3 + T)}\right) \quad (3)$$

The optimal value of VPD for a particular crop depends on its growth stage. In early stages of foliage growth, typically a lower VPD is preferred where as at later stages of growth such as flowering, a higher VPD is more appropriate. Figure 14 shows the values of daytime VPD calculated in mbar from 8 am to 5 pm. With a mean value of 2.36 mbar, standard deviation of 1.29, and a variance of 1.67, the high variability in VPD signifies real-time monitoring of environmental conditions for effective irrigation management. It is reflective of a highly dynamic environment where the interplay of relative humidity and temperature creates varying growing conditions. This is evident in Fig. 14 where VPD values remain lower in the early hours of the day and rise during the afternoon hours, peaking at 3 pm. This means, as the temperature rises, the humidity decreases, air gets relatively drier, resulting in less vapor pressure on the plant leaves, allowing relatively higher transpiration rate and translating into higher irrigation frequency over a longer period. This can be ascertained with larger datasets created over multiple crop cycles, which is one of the limitations of this work and an area for further improvement, as data was collected over a relatively shorter period of time.

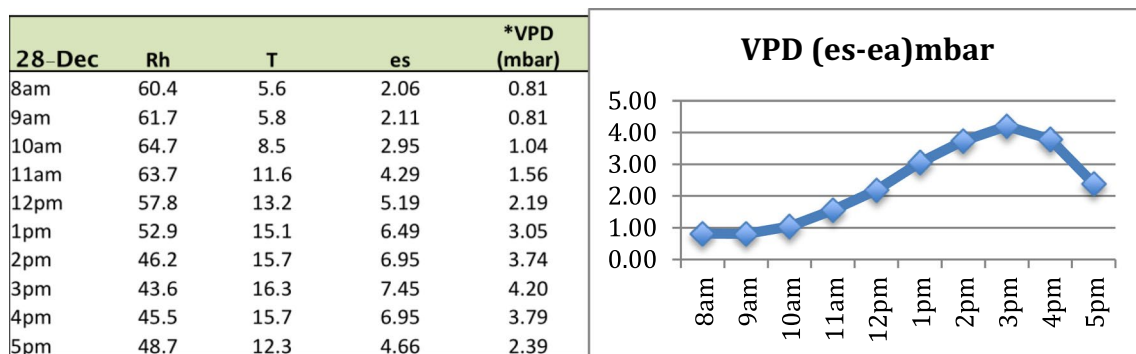


Fig. 14 VPD Calculation for the greenhouse setup

7 Conclusion

In recent years, the devastating effects of climate change appeared in more pronounced ways than ever in the form of extreme weather swings. The worst droughts seen in decades have caused acute water shortages, resulting in severe crop stress and below-par yields. In order to deal with such situations, a holistic approach is required to adopt sustainable agricultural practices that enable efficient and optimal use of resources. Technology has a key role to play in enabling efficient and sustainable agricultural practices and recent advancements in IoT, AI, Cloud and Edge Computing, Big Data Analytics and UAVs are pivotal in this cause.

In this work, we reviewed smart agriculture enabling technologies and identified key challenges and opportunities for sustainable agriculture practices in resource-constrained environments. We proposed a conceptual framework for smart agriculture consisting of soft and hard elements. The softer elements of this framework include policy & regulations, capacity building and smart agriculture ecosystem development. The hard elements include the technology solutions such as IoT sensors, network storage, computing, and communication technologies, and visualization, analysis and decision-making tools. In order to create precise and accurate monitoring and analysis capabilities, site-specific data is required. For that a comprehensive strategy has to be devised to come up with affordable IoT solutions that meet the local requirements. We proposed low-cost open-source end-to-end IoT solution for smart agriculture consisting of IoT sensor nodes, gateway and cloud services to enable real-time monitoring and data collection. The efficacy of this system was demonstrated by implementing it in a low-cost bamboo greenhouse structure. In addition to real-time raw data for monitoring of environmental and soil conditions, we extracted additional features from primary data, and derived two crop growth indicative parameters, namely GDD and VPD. The real-time monitoring of soil conditions coupled with VPD helped determine precise irrigation requirements as against fixed scheduled irrigation. The observed water saving was approximately 50% over a two-week period, although a longer run would be required to get a more accurate estimation of water conservation over the crop lifecycle. Similarly, GDD helped estimate the crop maturity date instead of relying on fixed calendar days, which is crucial for planning purposes. Since we chose not to perform any preprocessing of data to be able to qualitatively assess sensor performance, we believe that future improvements may incorporate lightweight preprocessing algorithms running on the Edge gateway for removing jitters in data, replacing missing values, detecting outliers, and extracting features—issues frequently encountered with low-cost sensors. Similarly, data collected over longer periods will help create larger datasets, which can be used to perform more meaningful statistical analysis for better insights.

Technology can effectively protect agriculture against the adverse effects of climate change. IoT and AI are key enabling technologies in this context. Their application in resource-constrained environments is a challenging task that requires a systematic approach. The framework proposed in this work is an attempt to address this and we encourage stakeholders, particularly researchers and practitioners to further explore, expand, develop, implement, and experiment with it. A few recommendations in this regard are summarized below.

7.1 Recommendations

In order to encourage application of technology in agriculture in resource-constrained environments, an interdisciplinary approach is required, starting at the policy level. This has to be followed by a well thought through strategy for extensive research and development (R&D) efforts, undertaken through public–private partnerships, involving tech industry, academia, and agriculture businesses. The output of R&D initiatives should result in a startup ecosystem providing products, solutions and services for smart agriculture. The bootstrapping and subsequent strengthening of entrepreneurial ecosystem through incubations and venture capital (VC) funding can transform the conventional and depleting arable agriculture sector, resulting in highly efficient farming practices. Effective and targeted incentives in the form of credits, subsidies, and seeding—creation of a few model farms with precision agriculture for demonstration—can significantly increase the uptake of smart farm solutions.

Effective utilization of existing infrastructure to educate, encourage, facilitate, and support farmers to adopt new technology initiatives and collaboration with other departments for data collection is paramount. New products and services for smart agriculture can be introduced systematically on top of these efforts, increasing the chances of success and uptake.

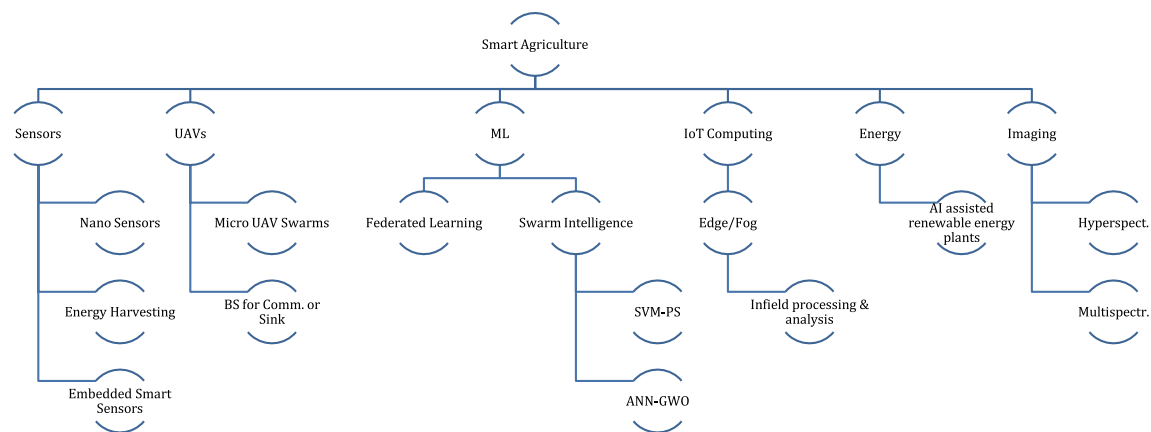


Fig. 15 Future research areas in smart agriculture

Data is the lynchpin on which the edifice of smart agriculture stands. In order to do any meaningful and intelligent analysis and decision-making for agricultural operations, a clean relevant set of site-specific data is required. Obtaining quality datasets with sufficient quantity of features and instances remains a challenge, as it takes significant effort to collect site-specific data, preprocess it to remove errors, duplicates, outliers, handle missing values and imbalances in data, and add labels and annotations. This is an area where much work is required to enable creation of datasets based on well-defined guidelines, standards and practices.

Open-source platforms need to be developed to facilitate Farm-as-a-Service (FaaS) model for providing farm management services. Mobile applications using location and farm specific information should be developed in regional languages to communicate with farmers effectively for annunciations, alerts, recommendations, predictions etc. Focus should also be on developing and deploying self-sustainable smart farm systems for small farms in remote areas that can provide monitoring, control and inference services in the absence of cellular and internet coverage. Low cost smart sensors, Edge computing and ML for embedded and mobile devices are the prerequisites for such solutions. An innovative approach in this regard can be the provision of UAV services to small farmers who cannot afford or have the technical skills to operate these. These UAVs can provide crucial crop canopy temperature, water requirement and vegetative health information services through multispectral, hyperspectral and near-infrared imaging. Services using remote sensing through satellites and GIS can provide area specific information.

8 Future research areas

Smart Agriculture is a rapidly evolving field. Smart IoT nodes with the capacity to perform data preprocessing and run ML algorithms for inferences, utilizing specially designed frameworks for embedded machine learning, such as TinyML, will remain a focus of current and future research. This creates new requirements not only for the sensor architecture but also for developing lightweight ML algorithms for resource-constrained devices. These developments will bring real-time monitoring and intelligent decision-making capabilities to the fields where these sensors will be installed for applications such as smart irrigation, fertilization, disease detection, and maintaining optimal growth conditions. Network bandwidth, latency, privacy, and security concerns will be adequately addressed by keeping all raw data onsite. Similarly, advances in compressed sensing and sensor data fusion promises efficient use of network and computing resources by reducing the amount of data generated and communicated in the IoT network. Energy harvesting for IoT sensors remains an area of great interest to increase the network lifetime and has potential for application of AI algorithms to create AI-assisted energy harvesting systems. The shift in focus from centralized to decentralized computing approaches, to address privacy and security concerns among others, drives the Edge computing research. Scalability remains a challenge in this area. Distributed ML training strategies such as Federated Learning also strive to address these issues. UAVs have been utilized for data collection, monitoring and imagery using visual, multispectral, and NIR images. Micro-UAV swarm intelligence holds a promise to add a new dimension to UAVs application in smart agriculture. Potential applications include soil

and crop canopy monitoring for estimating irrigation and fertilization requirements, early weed and disease detection, and crop growth monitoring.

Data availability in remote and rural areas is sparse and remains a challenge. Interpolation of available geospatial environmental data using GIS models and ML algorithms to cover larger areas is required. This will enable creation of agricultural indices of interest for crop growth monitoring in areas where direct sources of data are not available. Real-time monitoring with computer vision and DL can address the challenge of timely detection of critical events, such as disease onset and assist in generating prompt responses. Green IoT for sustainable farming practices is creating a lot of buzz and intends to develop environmentally friendly IoT technology. Figure 15 summarizes the future research areas in smart agriculture discussed in this section.

Author contributions M.N. and M.I.K.B. conceptualized the research work, M.N. and M.I.K.B. devised methodology, carried out experimentation, collected and tabulated results, M.N. drafted the paper, M.I.K.B. supervised and reviewed the work. All authors have read and agreed to the manuscript.

Data availability Data is provided within the manuscript.

Declarations

Competing interests The authors declare no competing interests.

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